

Efficient Construction of Weakly-Connected Dominating Set for Clustering Wireless Ad Hoc Networks

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Abstract—In most of the proposed clustering algorithms for wireless ad hoc networks, the cluster-heads form a *dominating set* in the network topology. A variant of dominating set which is more suitable for cluster formation is the *weakly-connected dominating set* (WCDS). We propose an area based distributed algorithm for WCDS formation with time and message complexity $O(n)$. In this *Area* algorithm, we partition the wireless nodes into different areas, use some deterministic criteria to select the nodes for the WCDS in each area and adjust the area borders by adding additional nodes to the final WCDS. The effectiveness of our algorithm is confirmed through analysis and comprehensive simulation study.

Keywords—Wireless ad hoc networks; clustering; dominating set; weakly-connected dominating set; distributed algorithm

I. INTRODUCTION

Wireless ad hoc network shows great potential and importance in many situations because of its independence of a fixed infrastructure, its instant deployment and easy reconfiguration capabilities. However, a flat ad hoc network has poor scalability [1]. Theoretical analysis in [1] implies that even under the optimal circumstances, the throughput for each node declines rapidly toward zero while the number of nodes is increased. Thus, new methodologies are needed for building large scale ad hoc networks. Among the solutions proposed for scaling down large-sized networks, *clustering* is one of the most investigated approaches [2-5]. It has been pointed out that “The most basic clustering that has been studied in the context of ad hoc networks is based on dominating sets” [6]. In general, a *dominating set* (DS) of a graph $G = (V, E)$ is a subset $V' \subseteq V$ such that each node in $V - V'$ is adjacent to at least one node in V' , and a *connected dominating set* (CDS) is a dominating set whose induced sub-graph is connected. Unfortunately, the dominating set and connected dominating set problems have been shown to be NP-Complete [7, 8]. Even for a unit disk graph (UDG) [9], the problem of finding a minimum CDS (MCDS) is still NP-Complete [10].

In the general approach of clustering ad hoc networks, nodes are first added to the dominating set according to some local condition. The nodes in the dominating set are called *cluster-heads* (or *dominator*). Different heuristics approaches exist for electing cluster-heads. Characteristics upon which these heuristics are based include *node ID*, *node degree*, *remaining battery life* or *node mobility*. Unfortunately, the cluster-heads in existing clustering algorithms usually form a DS. After some gateway nodes are selected [3], cluster-heads and gateways can sometimes construct a CDS. Although a CDS

provides an obvious routing for messages, the connectivity requirement causes the size of CDS to be rather large. Relaxing this requirement can reduce the size effectively. In a graph $G = (V, E)$, the sub-graph weakly induced by $S (S \subseteq V)$ is the graph $S_w = (N[S], E \cap (N[S] \times S))$, where $N[S]$ includes the nodes in S and all of their one-hop neighbors. The edges of S_w are all edges of G that have at least one end point in S . The subset S is a weakly-connected dominating set, if S is a dominating set and S_w is connected. Dunbar et al studied weakly-connected domination in graphs and showed that the problem of finding a minimum size WCDS in a given graph is NP-Complete [11].

This paper presents an area based distributed algorithm for WCDS construction in ad hoc networks with constant approximation ratio, linear time and message complexity. The *Area* algorithm partitions the wireless nodes into different areas, constructs WCDS in each area and adjusts the area borders by adding additional nodes to the WCDS. Compared with previous algorithms, the contribution of this scheme mainly lies in that we use the *Area* concept to guarantee the weak-connectivity by the selected cluster-heads in each area. Therefore, the approximation ratio of WCDS in each area is reduced from 122.5 to 5 and the size of WCDS for the whole graph is significantly decreased. The effectiveness of our approach is confirmed through analysis and comprehensive simulation study.

The rest of the paper is organized as follows. Section II introduces the related work. In Section III, we present our distributed WCDS formation algorithm and give the performance analysis. Section IV presents the simulation results. We point out future directions and summarize major results in Section V.

II. RELATED WORK

A. Connected Dominating Set

Das et al proposed a MCDS based routing algorithm for ad hoc networks [12]. This algorithm is a distributed version of Guha and Khuller’s approximation algorithm to calculate CDS [13]. The algorithm proposed by Wu and Li first finds a CDS and then prunes certain redundant nodes from the CDS [14]. Stojmenovic et al also presented a distributed construction of CDS in the context of clustering and broadcasting [15]. For the constant approximation ratio schemes in UDG, Wan et al’s scheme has performance ratio of 8, message complexity of $O(n \log n)$ and time complexity of $O(n)$ [16]. The scheme finds a *maximal independent set* (MIS) as a dominating set. Dubhashi et al proposed a polylogarithmic-time distributed

algorithm that finds an approximation to the MCDS [17]. Recently, Basagni et al gave a comparative analysis for CDS computation [18]. A nice literature review of connected dominating set formation for clustering ad hoc networks can be found in [18].

B. Weakly-Connected Dominating Set

WCDS was first proposed for clustering ad hoc networks by Chen and Liestman [19]. The distributed approximation algorithm in [19] is also inspired by Guha and Khuller's centralized algorithm [13] for CDS formation. However, this distributed algorithm is not localized because it uses some central coordinators to direct the algorithm execution. To address the problem of non-localized computation, Chen and Liestman also proposed a zonal algorithm [20], hereafter referred to as Zonal. In this approach, they divided the graph into regions. The root of each region controls the region size by picking a value x . Alzoubi et al presented two distributed algorithms for finding a WCDS in $O(n)$ time [21]. The first algorithm has an approximation ratio of 5 and requires $O(n \log n)$ messages. The second algorithm first constructs a MIS and then modifies the MIS by adding one intermediate node between each pair of three-hop away dominators. It has a larger approximation ratio (122.5), but requires only $O(n)$ messages.

III. AREA ALGORITHM FOR WCDS FORMATION

A. Network Assumptions and Preliminaries

Like many existing approaches, we do not deal with the issue on how the messages use a shared channel to avoid contention and collision. We assume that an ad hoc network comprises a group of wireless nodes communicating through a common broadcast channel using omni-directional antennas with the same transmission range. That is, the corresponding topology graph is UDG. Each node has a unique ID (e.g., IP address) and it also knows the ID of its neighbors, which can be achieved through periodically broadcasting "HELLO" messages by each node. In the rest of this subsection, we give some well-known preliminaries that will be used in this paper.

Preliminary 1: By building a DS through MIS construction, for every node u , the number of dominators inside the disk centered at u with radius k -units is bounded by a constant l_k .

Proof: Alzoubi et al gave a proof through calculation that $l_k < (2k + 1)^2 - 1$ [21]. When $k = 2, 3$, we have $l_k = 23, 47$. Let D be the maximum density of the packing of n equal small circles with radius r in a large circle with radius R . Thus, $D = n\pi r^2 / \pi R^2 = nr^2 / R^2$ [22]. The upper bound of D is also given in [23] as

$$D \leq \frac{n}{\left[1 - \frac{\sqrt{3}}{2} + \sqrt{\frac{3}{4} + \frac{2\sqrt{3}}{\pi}(n-1)}\right]^2} \quad (1)$$

When $k = 2$, $R = 2r$, $D = nr^2 / (2r)^2 = n/4$. From (1), we can calculate a smaller $l_2 = n = 21$. Similarly, we can get $l_3 = 42$. ■

Preliminary 2: Let G be a unit disk graph and OPT be the size of a minimum WCDS for G , then the size of any maximal independent set for G is at most $5 \times OPT$.

The proof of this preliminary bounds the size of any MIS in G and can be found in [21]. ■

B. Area Algorithm

Our main objective is to design a message-efficient localized algorithm which produces a smaller WCDS. The proposed algorithm follows the general clustering approach [2-5]. At first, we construct a DS through MIS, and then we add more nodes to this DS to make it weakly-connected. Compared with existing approaches [19-21], the novelty of this algorithm is that during the DS formation, the wireless nodes are divided into several areas naturally and the dominators in each area form a WCDS of that area automatically. What we need to do after that is only adjusting the area borders to make these areas weakly-connected. It is the reason why we introduce the *Area* concept! Moreover, the root of each area does not need to control the area size and adjust the area borders.

For simplicity, in the following description of Area algorithm, we will use node degree as the deterministic criterion to select the cluster-heads. We stress that our algorithm can be easily extended to use other selection criteria, such as available energy, node mobility, or some combinations of them. Define the rank of node u to be an ordered pair of (δ_u, id_u) where δ_u is the node degree and id_u is the node ID of u . We say that a node u with rank (δ_u, id_u) has a higher order than a node v with rank (δ_v, id_v) if $\delta_u > \delta_v$, or $\delta_u = \delta_v$ and $id_u < id_v$. As mentioned above, the first two parts of Area algorithm, area partition and WCDS construction in each area, are naturally combined with each other through DS formation. We will illustrate these two parts in the following subsection.

1) *Area partition and WCDS formation in each area:* In the first phase, we partition the nodes into different areas and each area is supposed to have a unique area ID. Thus, each node is also assigned an area ID to indicate which area it belongs to (Details about how nodes obtain their area ID will be discussed later). Each node is in one of the four states: *unmarked*, *cluster-head*, *cluster-member* and *additional-cluster-head*. First, a node u with the highest rank among its unmarked one-hop neighbors becomes a *cluster-head* and broadcasts a CH (stands for cluster-head) message to all its neighbors. Note that such node does exist in the beginning. After receiving a CH message, a node, say v , becomes a *cluster-member* if its current state is unmarked. If it is the first time that v receives a CH message, v will broadcast a CM (stands for cluster-member) message to all its neighbors. The same procedure is repeated among the remaining nodes, until each node becomes either a cluster-head or cluster-member.

In a word, a node u can become a cluster-head if and only if: (1). Node u has the highest rank among all its one-hop neighbors; or (2). Node u has neighbors with higher rank than itself, but these neighbors already become the cluster-members of other clusters. We call a cluster-head that satisfies condition (1) as *root cluster-head* and that satisfies condition (2) as *non-root cluster-head*. Obviously, the condition that a node becomes a non-root cluster-head is indirectly affected by some root cluster-head.

To partition the graph, the node ID of a root cluster-head automatically becomes the ID of the corresponding area

centered at this root cluster-head. We add an item, *Area ID*, into the CH message to indicate the area that the cluster-head belongs to. When an unmarked node receives the first CH message, it becomes a cluster-member of the area indicated in this message. Each cluster-member also inserts its area ID into the CM message that it broadcasts to all its neighbors. From its neighboring cluster-members, every non-root cluster-head knows to which area it belongs. If neighboring cluster-members have different area IDs, the non-root cluster-head can arbitrarily select one area to join. The nodes with the same area ID form an area eventually and the graph is divided into different areas through a sweep of the network spreading outwards from the root cluster-heads. Moreover, the cluster-heads in each area form a WCDS of that area.

2) *Adjusting area borders*: After the partition, each cluster-head has at least one two-hop away neighboring cluster-head if there is any other one existing in the same area. It is the reason that the cluster-heads in each area form a WCDS of the induced sub-graph for each area. Combining the WCDS for each area does not necessarily generate a WCDS for the entire graph because at the area borders there may exist some pairs of three-hop away neighboring cluster-heads located in different areas. In order to obtain a WCDS for the whole graph, additional nodes at the area borders should be added into the final WCDS.

To adjust the area borders, we need to know which cluster-heads and cluster-members are in the area borders. A cluster-member is a *border cluster-member* if it has a neighbor with different area ID (which can be known from CM message). For example, in Fig. 1, nodes p , q , x and z are all border cluster-members. Similarly, a cluster-head is a *border cluster-head* if it has a two or three-hop away neighboring cluster-head with different area ID. To make border cluster-heads know their two-hop away neighboring cluster-heads, border cluster-members need to broadcast a ONE-HOP-CH message that includes all their known one-hop away cluster-heads and their area ID. After this, border cluster-members will also know their two-hop away neighboring cluster-heads. Then to make border cluster-heads know their three-hop away cluster-heads, border cluster-members need to broadcast another TWO-HOP-CH message that includes all their known two-hop away cluster-heads and their area ID.

The border cluster-head will then select some cluster-members to become additional-cluster-head from its own local view. For clarity, we still use Fig. 1 to illustrate this procedure. Take node u as an example. From its local view, A_1 and A_2 , A_1 and A_3 are all weakly-connected, because it has two-hop away neighboring cluster-heads in A_2 and A_3 . But from node v 's local view, A_2 and A_3 are not weakly-connected, because node v only has a three-hop away neighboring cluster-head in A_3 . In order to adjust the border of A_2 and A_3 , node v will notify node q to become additional-cluster-head. To avoid collision, in this pair of three-hop away neighboring cluster-heads, only cluster-head with lower ID need to adjust the border. Thus border cluster-head w will take no action. Moreover, if there are multiple paths connecting a pair of three-hop away neighboring cluster-heads in different areas, the cluster-member with higher rank has the priority to be selected as additional-cluster-head.

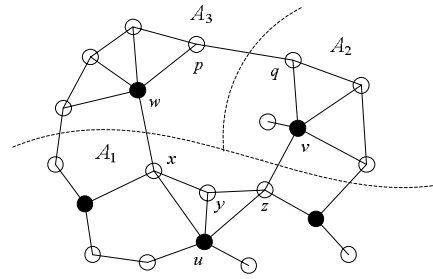


Figure 1. Adjusting the area borders

3) *Mobility*: For the limited space, we do not discuss the maintenance of WCDS during the network lifetime. We note that the technique proposed in [24] can be easily integrated into our scheme to deal with the node mobility.

C. Performance Analysis

In this subsection, we show the correctness, analyze the time and message complexity of Area algorithm, and also give the approximation ratio of it.

Theorem 1. *All the cluster-heads and additional-cluster-heads selected by Area algorithm form a WCDS.*

Proof: If there is any other cluster-head existing in the same area, it can be easily proved that each cluster-head has at least one *two-hop* away neighboring cluster-head in the area it belongs to. Therefore, all the cluster-heads in the same area form a WCDS of the area. Of course, the WCDS is also a MIS of the area. Note that we have adjusted the area borders to guarantee the adjacent areas to be weakly-connected, thus, proved the theorem. ■

Theorem 2. *Area algorithm has both time and message complexity of $O(n)$.*

Proof: The time complexity of the Area algorithm is bounded by $O(n)$. The worst case occurs when all nodes are distributed in a line and in either ascending or descending order of their ranks. Since each node sends a constant number of messages, the total number of messages is $O(n)$. Thus, both the time and message complexity of the proposed Area algorithm are $O(n)$. ■

Theorem 3. *Let G be a unit disk graph and OPT be the size of a minimum WCDS for G , then the size of WCDS constructed by Area algorithm is within a constant approximation ratio of OPT .*

Proof: From Preliminary 2, we know that the size of MIS is at most $5 \times OPT$. The cluster-heads in each area form a WCDS of the induced sub-graph for each area. Let OPT' be the size of a minimum WCDS for the induced sub-graph of an area. Therefore, the size of WCDS we construct for the induced sub-graph of an area is at most $5 \times OPT'$ (because it is also a MIS). But to adjust the area borders, we add more nodes to the WCDS for the entire graph. Since each pair of border cluster-heads introduces at most one node into the WCDS, from Preliminary 1 and 2, the number of nodes in the WCDS is at most $(42/2 + 1) \times 5 \times OPT = 110 \times OPT$. ■

Theorem 3 only gives an upper bound of the size of WCDS constructed by Area algorithm. In fact, all the cluster-heads that are not at the area borders will not introduce other nodes

into the WCDS. Thus, in each area, the size of WCDS constructed by us is small. We will confirm the effectiveness of this algorithm through simulation study in the next section.

IV. SIMULATION

We compare the performance of two kinds of Area algorithm, Min ID (Area algorithm using node ID as the selection criterion) and Max Degree (Area algorithm using node degree as the selection criterion), with Zonal algorithm [20] (with region size parameter x set to 20 and 100) and the second algorithm in [21]. The first algorithm in [21] was implemented by first electing a leader r among the nodes. But, distributed leader election is a non local task that requires coordination and extremely message expensive in practice [18]. Thus, we do not consider it in the following performance evaluation. Hereafter, we call the second algorithm in [21] AWF algorithm (so named here for the initials of the authors' last names). We use a custom event-driven simulator to simulate wireless ad hoc networks. To measure the performance of the algorithms, we compare the size of WCDS constructed by different algorithms. The central algorithm [19], hereafter referred to as Central, is also considered as baseline for comparison.

A. Size of WCDS

First parameter is the size of WCDS constructed by these algorithms. We prefer smaller values of this parameter because it can reduce the routing overhead when this hierarchy is applied to route packets. In our simulation, a given number of nodes (ranging from 60 to 200 with increment step of 20 and from 200 to 1000 with increment step of 100, respectively) were randomly and uniformly distributed in a square simulation area of size 100 by 100 units. Each node has a fixed transmission range r ($r = 15, 30$ units, respectively). All the simulation results presented in this section were obtained by running these algorithms on 300 connected graphs. In fact, the density μ of graph can be calculated as $\mu(r) = n\pi r^2/A$, where n is the number of nodes in the graph, A is the area of the graph and r is the transmission range. Basically, $\mu(r)$ gives the average number of nodes within the transmission radius of each node. This allows us to test these algorithms with increasing network density from $\mu(r) = 4$ (sparse network) to $\mu(r) = 283$ (very dense network).

Figs. 2(a) and 2(b) show the simulation results when the node's transmission range is 15 units. From the figures, we found that when more nodes join the network, the number of nodes in the WCDS increases because of the increase of network density. Central performs better than all other distributed algorithms. In general, as the number of nodes increases, the gap between the curves for Min ID and Max Degree and the curves for other distributed algorithms becomes significant, thus, in these algorithms, the performance of our Area algorithm is the best. Among these algorithms, Zonal ($x = 20$) tends to construct the largest WCDS in dense networks. When $n = 1000$ and $r = 15$, the size of WCDS generated by Zonal ($x = 20$) is nearly ten times larger than that constructed by Max Degree. More detailed analysis and experiments about the size control parameter x of Zonal algorithm can be found in [20]. Generally, when x is small, the algorithm finishes quickly with a large WCDS. When it is

large, it behaves more like a non-localized algorithm and generates smaller WCDS. To show the difference of Central, AWF, Min ID and Max Degree, in Fig. 2(b) we omit the two curves for Zonal.

Figs. 3(a) and 3(b) show the results when the node's transmission range is set as 30 units and the number of nodes in the networks ranges from 60 to 200 and from 200 to 1000, respectively. For the reason mentioned above, we still omit the two curves for Zonal. When the transmission range increases, as more nodes may be connected, the network becomes denser if the number of nodes is fixed. In this case, the size of WCDS only increases slightly as the size of the network increases. When the number of nodes in the network reaches 1000, the number of nodes in the WCDS constructed by the Area algorithm is only about 50% of that constructed by AWF. The reason our algorithm always outperforms AWF is that for each pair of three-hop away cluster-heads AWF adds one additional node to the WCDS. Whereas our algorithm only "weakly-connects" two three-hop away cluster-heads in different areas. Comparing Figs. 2(a) and 2(b) with Figs. 3(a) and 3(b), we find that increasing the node's transmission range can increase the coverage area of each node and, therefore, increase the density of the network, which leads to a smaller size of the WCDS.

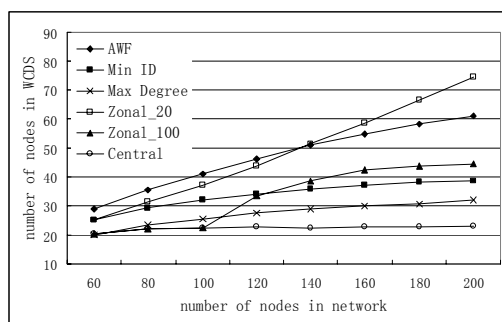
We also compare the message overhead of these algorithms. For the limited space, we show the results for communication overhead in [25].

V. CONCLUSION AND FUTURE WORK

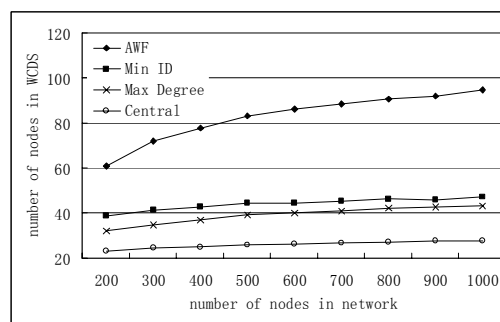
We have proposed an efficient area based distributed algorithm for weakly-connected dominating set construction in ad hoc networks. The time and message complexity of the Area algorithm are both $O(n)$. In this algorithm, we partition the wireless nodes into different areas, construct the WCDS in each area and adjust the area borders by adding more nodes to the final WCDS. We confirm the effectiveness of our algorithm through a comprehensive simulation study on both sparse and dense networks, compared with previous well-known algorithms. From the simulation results, we have observed that Area algorithm always outperforms the other distributed algorithms regardless of the size and density of the network in terms of the size of WCDS. Considering other graphs than UDG and more cluster-head selection criteria are our current issues of research.

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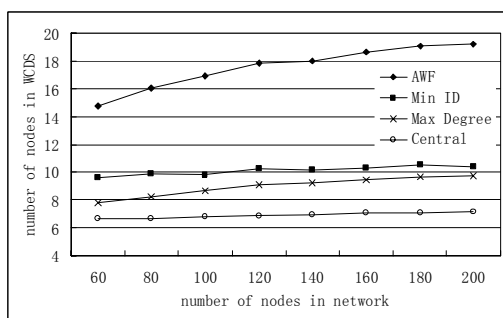


(a) # of nodes in WCDS ($n \in [60, 200]$)

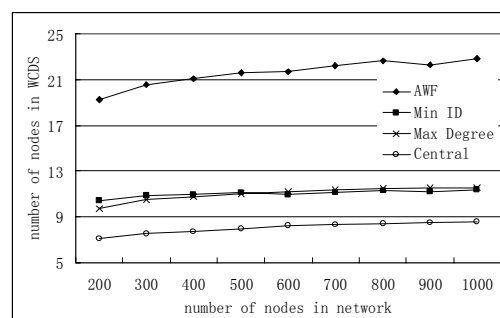


(b) # of nodes in WCDS ($n \in [200, 1000]$)

Figure 2. The number of nodes in WCDS when r is 15 units



(a) n ranges from 60 to 200



(b) n ranges from 200 to 1000

Figure 3. The number of nodes in WCDS when r is 30 units

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